Puneeth Yashasvi Kashyap Apparasu

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Skilled professional with a strong academic background and over 3 years of experience in Python, Machine Learning, Generative AI, and software development. Specializing in large language models (LLMs), fine-tuning, and RAGs, with 2+ years of dedicated expertise in these areas. Effective in supporting organizational goals through strategic application of machine learning and Generative AI techniques.

SKILLS

Languages: Python, Flask, R, C/C++, SQL
Frame Works: Django, OpenCV, TensorFlow, Keras, Pytorch, CUDA, OpenMP, Django-rest, React.js, Langchain, Llama Index, Celery, Pandas, NumPy, SciPy, Sklearn
Data Engineering and Analytics: MongoDB, MySQL, PostgreSQL, Pinecone**,** Azure storage service, RDS, ETL, Athena, Kinesis, S3, Spark, Data Analytics
ML/AI: Machine Learning, Deep Learning, ML Pipeline orchestration, Generative Adversarial Networks (GANs), Computer Vision, Natural language processing (NLP), Large Language Models (LLMs), LLM Finetuning, Quantization, Knowledge Distillation, Retrieval Augmented Generation (RAG), Transformers, PEFT, OpenAI, Llama, Gemini pro, Huggingface, Bedrock, Sagemaker, Distributed LLM training, RLHF
Tools and Platforms: Git & GitHub, Kubernetes, Docker, OpenShift, Azure Devops, AWS, Azure
Software Development: Backend Development, API optimization, parallelizing with Concurrent Futures

EDUCATION

Master of Science in Machine Learning
Stevens Institute of Technology, NJ
Jan 2024, CGPA: 3.8/4.0

B. Tech in CS & Engineering
Indian Institute of Information Technology, Sri City, IN
2022, GPA: 3.56/4.0

CERTIFICATIONS

Certified Tensorflow Developer (exp: 05/2027)
Machine Learning
MLOps specialization
Deep Learning Specialization
Build Better Generative Adversarial Networks
Quantization Fundamentals
Finetuning Large Language Models
Langchain Chat with your data
AWS Generative AI with LLMs
Functions, Tools and Agents with Langchain
IBM data engineering
Oracle Certified Generative AI Professional

PUBLICATIONS

Synthetic Sar Image Generation Using Generative Adversarial Networks: Intl. Research Journal of Modernization in Engineering Technology and Science, Vol: 03/Issue: 10/October-2021.

Developed new DCGAN architecture that generally works well on SAR data domain. Generated synthetic SAR images to reduce resources and cost by avoiding traditional methods of SAR imaging.

Emotion Recognition from Brain Waves While Listening to Music: 13th Intl. conference on Intelligent human computer interaction, published in Springer LNCS.

Used Deep learning to classify and recommend music based on emotion pertaining to the brain waves. Achieved great accuracies by ensembling various models compared to existing literature.

EXPERIENCE HIGHLIGHTS

Machine Learning Engineer – Smartmirror LLC, Houston, Texas — Sept 2024 – Present

- **Pioneered Generative AI Adoption:** Developed proprietary algorithms, cutting external API costs by 80%.
- **Scaled ML Solutions:** Designed and deployed scalable solutions on AWS (Lambda, EC2, S3) to serve 150,000 users.
- **Optimized Background Processing:** Engineered an AWS-hosted Flask app with in-memory job tracking and custom Gunicorn settings for efficient LLM processing.
- **Personalized Makeup Recommendations:** Created a system on AWS Lambda using GPT-4, SerpAPI, and YouTube API, delivering customized product Recommendations and tutorials.
- **Patient Support via Vision Models:** Built a video processing tool to analyze asthma inhaler usage, improving patient support through model-driven insights.

Software Engineer (Gen AI) – Blujin Corp., New Jersey — June 2023 – Sept 2024

- Developed end-to-end machine learning services for the company's application, utilizing Python, Django REST, OpenAI's API, and MongoDB.
- Achieved automation and optimal flow with Celery schedulers, enhanced performance through advanced prompt engineering, multi-threading, and API optimization.
- Successfully finetuned large language models (LLMs) using PyTorch, Llama. Built and deployed an application leveraging OpenAI and Azure for scalable cloud computing.
- Utilized Pandas for efficient data preprocessing and feature engineering. Created custom queues in MongoDB to streamline the processing of multiple user and client processes.

Data Scientist – Retail Pulse, Bengaluru, India — Jan 2022 - Jul 2022

- Provided technical expertise in collecting and organizing large datasets using Pandas and PySpark for efficient analysis.
- Conducted exploratory data analysis and implemented statistical techniques to extract valuable insights.
- Applied supervised and unsupervised learning techniques using to identify, analyze, and interpret trends.
- Supported the development and evaluation of machine learning models for various business applications.
- Designed, implemented, and evaluated innovative computer vision algorithms.
- Achieved significant improvements in task automation, model performance, and data processing, leveraging Docker, Kubernetes, and AWS for scalable deployment and optimal performance.

AI Engineer – 9I Web Solutions, Hyderabad, India — Dec 2020 – Dec 2021

- Collaborated with cross-functional teams to integrate AI technologies, resulting in a 20% improvement in client business processes and decision-making efficiency.
- Applied deep learning techniques to develop intelligent chatbots, improving customer service interactions and reducing response times by 30%.
- Worked on Data Annotation, object detection and semantic segmentation, delivering AI solutions that aligned with client strategic goals and provided measurable value.

NOTABLE PROJECTS

- **Detoxifying LLM completions:** fine-tuned a FLAN-T5 model for less-toxic summaries using reinforcement learning (PPO), PEFT and Meta AI's hate speech reward model. Gained valuable insights gained into reinforcement learning techniques and model refinement.

- **Cryptocurrency Conversational Chatbot Project:** Utilized prompt engineering, Lang chain tools and Agents to develop a chatbot capable of accessing knowledge stored in a vector database powered by Pinecone. Integrated SERPAPI to provide comprehensive answers to user queries. Implemented memory functionality within the bot, enabling it to access tools and resources to deliver accurate responses tailored to user inquiries.

- **Event Association Analysis on Covid-19 Vaccine Adverse Reactions**: Conducted events association analysis on the VAERS dataset, leveraging natural language processing (NLP), transformers, and large language models (LLMs). Applied advanced techniques to identify discernible patterns of symptom reporting, achieving significant confidence levels of up to 85% on certain events.